Daniele Filaretti

Software Engineer

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Work Experience

Apr. 2025 to Oct 2026 — **Senior R&D Engineer**, *Veridise*, (Remote).
Developing static analysis tools for detecting security vulnerabilities in Solidity smart contracts in the DeFi space (C++, MLIR).

Jun. 2020 to Mar 2025 — **C++ Developer (contract)**, *Loopmasters/Beatport*, (UK/Remote).
Contributed to the development of the Loopcloud desktop app, introducing new major features, an extensive UI rework, bug fixes and workflow improvements.

Aug. 2021 Jun 2022 — **C++ Developer (contract)**, *K-Devices*, (France/Remote).
Developed every aspect of the "TATAT" MIDI plugin based on the specification from the client, including vectorised UI, low-level MIDI engine, MIDI export, installers etc.

Jul. 2019 to Jun. 2020 — **Clojure(script) developer (full-stack)**, *Yapster*, (London, UK).
Helped developing Yapster - a messaging app for the workplace; contributing new features across both frontend and backend as well as bug-fixes.

Mar. 2018 to Feb 2019 — **R&D Engineer / Haskell developer**, *Runtime Verification*, (USA/Remote).
Contributed to several high-profile open-source projects in the blockchain, cryptocurrency and formal methods spaces (main tools: C++, Haskell, the K framework).

Sep. 2017 to Mar 2018 — **C#/Unity3D developer**, *Synergation Limited*, (London, UK).
Lead the development of DIMMAND, a cross-platform digital solution for detecting literacy difficulties & strengths in children.

Jan. 2016 to Aug 2017 — **C#/.NET embedded Software Engineer**, *Solid State Logic*, (Oxford, UK).
Helped develop and maintain the software powering Solid State Logic's high-end digital consoles in both the Broadcast (System T) and Live (L200, L350 etc.) product families.

Oct. 2011 to Jul. 2015 — **PhD Researcher**, *EPSRC (Grant EP/1004246/1), Imperial College London (UK).*
Developed formal specification of the PHP language and a prototype PHP bug finding tool.

Apr. 2009 to Dec. 2009 — **Sound Technician, Sound Designer (internship)**, *AGON, Milan (Italy).*
Helped sampling sound from 1500 streets and turning it into ambient music (*Audioscan Milan*)

Oct. 2008 to Feb. 2009 — **Software Engineer (internship)**, *University of Milan (Italy).*
Added syntax highlighting to the open source text editor *ne* (`http://ne.di.unimi.it/`).

Education

2011 - 2015 — **PhD (Computing)**, *Imperial College London*, *Funded by EPSRC scholarship.*

2009 - 2011 — **MSc (Information Technology)**, *University of Milan*, 110/110 (with honors).

2006 - 2009 — **BSc (Information Technology)**, *University of Milan*, 105/110.

Continuing Education & Certificates

Jan 2020 — **Advanced Audio Plugin Development**, *Output (on Kadenze).*

Jan 2017 — **Digital Signal Processing**, *École Polytechnique Fédérale de Lausanne (on Coursera).*